

Notes for ECE 50877 - Introduction to Applied Cryptography

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Course Description

Cryptography is an essential component of securing communication and data in a wide range of applications. This course explores fundamental building blocks in cryptography such as ciphers, hash functions, and digital signatures. We will then demonstrate how these building blocks are used practically to secure communications and systems. We will also learn about the power of advanced cryptography to enable complex functionalities with rigorous security guarantees. We will cover fundamental secure multiparty computation protocols, homomorphic encryption, and blockchains, and review their application in building privacy-preserving machine learning. We will also explore social and ethical issues in developing and deploying cryptographic systems.

Basis of Cryptography

Consider the archetypical case in cryptography. Alice wants to send a message to Bob, and wants the contents to remain secret from Vincent. Alice encrypts her message with a secret key s .

An encryption scheme (Gen, Enc, Dec) is a triple of algorithms where

- Key generation $\text{Gen}(1^n)$ outputs a secret key k of length n (also called the security parameter).
- Encryption algorithm $\text{Enc}_k(m) = c$ outputs ciphertext c .
- Decryption algorithm $\text{Dec}_k(c) = m$ outputs plaintext m .

An encryption algorithm is defined over a message space M and a key space K . Requirements:

- Correctness: $\text{Dec}_k(\text{Enc}_k(m)) = m$.
- Security: definition with respect to adversaries.

The secret key k should be hidden from the adversary. The encryption and decryption algorithms themselves should be public; this is Kerckhoffs's principle. This is because the secrecy keys are easy to hide and replace in case of a breach, not so with algorithms. It is also good to have many eyes on an algorithm to expose security flaws.

The key space must be large enough such that a brute force attack is infeasible. Consider a Caesar cipher; the key space is only 26 long and a message so encrypted is quickly bruteforced.

Shannon's Treatment: messages come from some distribution, let M be a random variable for sampling the messages from the message space M . Distribution M is known to the adversary. This captures a priori information about the messages. The ciphertext $c = \text{Enc}_k(m)$ depends on m chosen according to M . k chosen randomly according to Gen. Enc may also use some randomness. Dec is deterministic. These induce a distribution C over the ciphertexts c . The adversary only observes c . Knowledge about m before observing the output of C is captured by M . Knowledge about m after observing the output of C is captured by $M|C$. An encryption scheme (Gen, Enc, Dec) is *Shannon secure* if for every probability distribution for M , every message $m \in M$, and every ciphertext $c \in C$: $\Pr[M = m|C = c] = \Pr[M = m]$.

An alternative definition of secrecy is *Perfect Secrecy*. An encryption scheme is perfectly secure for every pair of messages m_1 and $m_2 \in M$ and every ciphertext $c \in C$ if $\Pr[\text{Enc}_k(m_1) = c] = \Pr[\text{Enc}_k(m_2) = c]$.

Shannon had fingers in many cryptographic pies. Shannon's Theorem says that if (Gen, Enc, Dec) is a perfectly secret encryption scheme with message space M and key space K , then $|K| \geq |M|$. This means

that for an encryption scheme to be perfectly secret, the keys must be as long as the message. The challenge of modern cryptography is to make something just as good as perfect for practical purposes.